

**Exercise 2.21.** Prove that  $n$  is even if and only if  $n^2$  is even. To do this, here are the two things that you should prove:

- (a) If  $n$  is even, then  $n^2$  is even.
- (b) If  $n^2$  is even, then  $n$  is even.

Solution.

We want to prove

$$n \text{ is even} \iff n^2 \text{ is even}$$

This means we need to prove **both directions**.

**(a) If  $n$  is even, then  $n^2$  is even**

**Proof**

Suppose  $n$  is even.

By the definition of an even integer, there exists an integer  $k$  such that

$$n = 2k.$$

Square both sides:

$$n^2 = (2k)^2.$$

Therefore,

$$n^2 = 4k^2.$$

Rewrite this as

$$n^2 = 2(2k^2).$$

Since  $2k^2$  is an integer,  $n^2$  is divisible by 2.

Thus,

$$n^2 \text{ is even.}$$

**(b) If  $n^2$  is even, then  $n$  is even**

We can prove this using contrapositive.

The statement is

$$n^2 \text{ even} \Rightarrow n \text{ even.}$$

Its contrapositive is

$$n \text{ odd} \Rightarrow n^2 \text{ odd.}$$

If  $n$  is odd, then by definition there exists an integer  $k$  such that

$$n = 2k + 1.$$

Square both sides:

$$n^2 = (2k + 1)^2.$$

Expand:

$$n^2 = 4k^2 + 4k + 1.$$

Factor out 2 from the first two terms:

$$n^2 = 2(2k^2 + 2k) + 1.$$

Since  $2k^2 + 2k$  is an integer, this has the form

$$2m + 1,$$

which is the definition of an odd integer.

Therefore,

$$n \text{ odd} \Rightarrow n^2 \text{ odd.}$$

By the contrapositive,

$$\boxed{n^2 \text{ even} \Rightarrow n \text{ even.}}$$

**Therefore**

Since both directions have been proven,

$$\boxed{n \text{ is even if and only if } n^2 \text{ is even.}}$$

Or, using logical notation:

$$\boxed{n \text{ even} \iff n^2 \text{ even.}}$$